

07:00 Fuzo Matungu (student no. 2805001951)

07:30

08:00 Exercise 2: SQL Aggregates & operators

08:30

09:00 (1) SELECT DISTINCT department
09:30 FROM students ;

10:00

10:30	department
11:00	IT
11:30	HR
12:00	Finance

12:30

13:00

13:30 (2) SELECT department,
14:00 AVG(age) AS avg-age
14:30 FROM students
15:00 GROUP BY department ;

15:30

16:00	department	avg-age
16:30	IT	20,5
17:00	HR	22
17:30	Finance	23

18:00

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"The majority of men meet with failure because of their lack of persistence in creating new plans to take the place of those which fail."

Napoleon Hill

NOVEMBER 2025

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

DECEMBER 2025

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

07:00

(3)

SELECT department

07:30

COUNT(student_id) AS student_count

08:00

FROM students

08:30

GROUP BY department

09:00

HAVING student_count > 1;

09:30

10:00

department	studentcount
IT	2
HR	2

10:30

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12:00

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(4)

SELECT student_id,

13:00

name,

13:30

age,

14:00

department

14:30

FROM students

15:00

WHERE age BETWEEN 21 AND 23;

15:30

16:00

16:30

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eva	22	HR

17:00

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JANUARY 2026

S M T W T F S
 1 2 3
 4 5 6 7 8 9 10
 11 12 13 14 15 16 17
 18 19 20 21 22 23 24
 25 26 27 28 29 30 31

FEBRUARY 2026

S M T W T F S
 1 2 3 4 5 6 7
 8 9 10 11 12 13 14
 15 16 17 18 19 20 21
 22 23 24 25 26 27 28

"The harder you work, the harder it is to surrender."
 Vince Lombardi

07:00

⑤

SELECT student-id,

07:30

name,

08:00

age,

08:30

department

09:00

FROM student

09:30

WHERE department IN ('IT', 'HR') AND ^{age}~~AGE~~ > 21;

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11:00

student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR
8			

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⑥

SELECT department,

14:30

SUM(credits) AS total-credits

15:00

FROM courses

15:30

GROUP BY department

16:00

HAVING total-credits > 5;

16:30

17:00

17:30

department	total-credits
IT	11

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"We all need people who will give us feedback. That's how we improve."

Bill Gates

NOVEMBER 2025

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
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DECEMBER 2025

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

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⑥ SELECT course_id,
 course_name,
 department,
 credits
 FROM courses
 WHERE NOT credits = 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑧ SELECT course id,
 course_name,
 credits

from courses

~~WHERE~~ ORDER BY credits DESC
 LIMIT 3

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

07:00 (9) SELECT MAX (grade) AS max-grade,
 07:30 MIN (grade) AS min-grade,
 08:00 AVG (grade) AS avg-grade
 08:30 FROM enrollments
 09:00
 09:30

max-grade	min-grade	avg-grade
90	78	84,6

12:00
 12:30 (10) SELECT ~~COUNT~~ COUNT (enrollment-id) AS enrollment-count,
 13:00 course-id
 13:30 FROM enrollments
 14:00 GROUP BY course-id
 14:30

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

07:00

(11) SELECT Sum(salary) AS total-salary,
Sum(bonus) AS total-bonus,
department

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08:00

08:30

FROM salaries

09:00

GROUP BY department

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department	total-salary	total-bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

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(12) SELECT department,
Avg(salary) AS avg-salary

13:30

14:00

FROM salaries

14:30

GROUP BY department

15:00

HAVING avg_salary > 55000

15:30

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16:30

department	avg_salary
IT	61 000
Finance	70 000

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07:00 (13) SELECT employee_id,
 07:30 name,
 08:00 salary,
 08:30 bonus
 09:00 (salary + bonus) AS total_compensation
 09:30 FROM salaries
 10:00 WHERE salary + bonus > 60000 ;
 10:30

11:00					
11:30	employee_id	name	salary	bonus	total_compensation
12:00	1	Tom	60000	7000	67000
12:30	3	Spice	70000	6000	76000
13:00					

13:30

14:00

14:30 (14) SELECT department
 15:00 sum(budget) AS total_budget,
 15:30 avg(budget) AS avg_budget

16:00

16:30

17:00

17:30

18:00	department	avg_budget	total_budget
18:30	IT	125000	270000
19:00	Finance	80000	80000
19:30	Marketing		
20:00			

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SELECT *

FROM projects

WHERE NOT department = 'marketing'

budget between 5000 AND 12000;

Project-id	Project-name	Department	Budget
1	AI App	IT	27000
5	HR Portal	HR	7000